Personal AI for Shared Spatial Work

Grounding dialogue and continuing joint action in XR

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For discussion with Associate Professor Selen Türkay, QUT

Research vision and starting point

I want to develop personal AI that people can talk to while working: an assistant that understands the surrounding activity, contributes useful actions and remembers shared progress. XR offers a way for this collaboration to take place around visible objects, drawings and models. My doctoral starting point is how a person and an embodied AI establish what they are discussing, connect that understanding to action, and continue the work when intentions or spatial conditions change.

Consider preparing a small exhibition. I point to a panel and say, “Turn this towards the entrance, but keep that work visible.” I sketch a visitor route, walk around the model and change my mind. The assistant should distinguish a temporary sketch from an agreed route, show the proposed rotation, and update its suggestion when I move a neighbouring exhibit. On a later visit, it should help recover why we chose the layout and reconsider what still fits. Architectural and immersive exhibition work provides a concrete setting for this broader personal AI vision.

Research context

Türkay and colleagues provide complementary starting points. Ngo, Harman and Türkay [1] explored how users perceive personalities of LLM-driven embodied conversational agents in VR. Their short exploratory study found discernible personality differences, with discrepancies between intended and perceived traits; interaction times did not differ significantly. Syiem and colleagues [2] studied human spatial dialogue supported by 2D and 3D drawings. Drawing modality changed communication strategies without establishing a performance advantage. Together, these works motivate asking how an embodied agent can participate in the spatial communication through which a task develops.

Contextual reference resolution and spatial co-creation are already established directions. GazePointAR uses gaze, pointing and conversation history [3]; GesPrompt integrates speech and co-speech gestures for XR operations [4]; VRCopilot supports layout co-creation and editable intermediate representations [5]. Eye2Eye combines shared attention, revisable memory and feedback [6]. My proposed contribution concerns a specific coordination problem: how a change to an expressed spatial relation should update dependent actions while preserving agreements that remain valid. The value of making these links visible and editable will be tested against a capable conversational baseline.

Research questions

RQ1 How do people establish and revise object references, spatial relations and intended effects while speaking, pointing, drawing and manipulating a shared model?

RQ2 When does an object-linked, editable working record improve collaboration with an embodied AI, compared with dialogue, highlights and action previews alone?

RQ3 When a goal or scene changes between visits, how can the assistant update affected actions while retaining useful agreements and the reasons behind them?

Interaction concept

The initial testbed will contain four to eight exhibits or display panels, a known entrance and a bounded set of spatial tools. Users will arrange, inspect and revise layouts in one Unity/OpenXR application. Speech and controller pointing will be the initial inputs, with a simple drawing tool for routes and relations. Gaze and unconstrained free-hand recognition are later options, contingent on equipment and formative findings.

The assistant can explain a relation, highlight a referent, draw a route or viewing line, compare alternatives and preview a move or rotation. A consistent voice and simple visible agent will provide an initial social presentation. The study will hold that presentation constant when testing coordination mechanisms; it will not assume that a humanoid appearance is necessary or that personality alone improves performance.

An editable working record

A candidate record links the object being discussed, the desired spatial relation, the evidence supporting that interpretation, and the next proposed action. For example, “keep this visible from the entrance” connects an exhibit, an observation point and a visibility condition. A viewing line and a concise explanation make that connection inspectable. The user can correct the referent, reposition the viewpoint or replace the intended relation through speech or direct manipulation. Only the relevant record appears beside the model, with an overview available on demand.

Entries distinguish explicit user statements, model assumptions, alternatives under discussion and superseded decisions. A sketch is initially treated as a candidate expression whose role may need clarification. Scene events and revisions identify which proposed actions need reconsideration. The record is a design hypothesis: formative work will determine which links users need and when exposing them creates unnecessary effort.

Connecting interpretation to action

Existing language models will interpret timestamped input and return structured proposals. Known scene objects, stable identifiers and a finite tool API will support highlight, draw, move, rotate, compare, explain and undo. Geometry checks will assess predefined conditions such as overlap and line of sight. Confirmed digital actions execute through the API. New object positions invalidate affected previews. Model claims, successful tool calls and satisfaction of the user’s intended relation will be logged separately. General visual reconstruction, autonomous physical manipulation and foundation-model training are outside the initial implementation.

Study 1 Understanding spatial coordination

A formative study with approximately 12–16 participants, with varied design experience, will examine paired layout and explanation tasks. Participants will speak, point, draw and inspect alternatives. Replay interviews will explore how an expression becomes an agreement, how sketches acquire meaning, and how people detect and repair different interpretations. A small disclosed Wizard-of-Oz iteration will then test candidate AI responses and record interfaces before model reliability dominates the experience.

Analysis will distinguish reference, viewpoint, relation, intention and execution problems, including ambiguous cases. The action-and-speech analysis in [2] offers a methodological starting point; ENA may complement qualitative analysis if the coded data and research question justify it. Outputs will be a coordination design space, representative change events and requirements for the prototype. These observations will inform rather than predetermine the working-record design.

Study 2 Testing shared understanding in action

A counterbalanced within-participant study will compare a conversational baseline with the same assistant plus the editable, object-linked working record. Both conditions will use the same model, task history, current scene, speech and pointing inputs, drawing tools, agent presentation, action API, highlights, previews, confirmation and undo. The baseline can revise its current interpretation through dialogue and inspectable summaries. This comparison tests the added record and editing workflow, rather than a capable AI against a deliberately weak assistant.

Matched tasks will include successful exchanges and controlled, reversible misunderstandings: a wrong referent, a relation interpreted from the wrong viewpoint, or an action based on a superseded goal. Identical error instances will be inserted into structured proposals at comparable task opportunities. They will be disclosed as a study manipulation and analysed separately from naturally occurring model errors. A separate end-to-end task will assess the actual running model without researcher correction.

Outcomes and analysis

The primary outcome will be successful repair before the first confirmed execution of the affected action: the interpretation and proposal satisfy an independently specified task requirement. Secondary outcomes will include repair time, clarification turns, unintended changes to agreed work, final constraint satisfaction and user effort. Task completion and quality will also be reported for exchanges without errors, so an effective repair interface is not assumed to improve ordinary work.

Approximately 24–30 participants is an initial planning range; a pilot and a prespecified precision target will determine the final sample. Task and condition order will be counterbalanced. Analysis will account for repeated participants and tasks, reporting effect estimates and uncertainty. Layout criteria will be defined independently; open design choices will be assessed against the participant's stated, revisable goal. Workload, perceived agency and social presence will complement behavioural measures. Model versions, latency, context budgets and failures will be recorded to support interpretation and replication.

Study 3 Continuing work after change

After the core system is stable, a repeated-use study will examine how the same assistant helps users resume and revise a spatial task. An initial planning range is 12–16 participants across three visits over two to three weeks. Each visit will involve arranging, inspecting and revisiting a bounded exhibition model. Changes will include a moved panel, a different visitor route or a revised visibility goal. Room-scale use will remain in the same tracked application; a physical counterpart with markers is an optional transfer step, not a prerequisite.

Both conditions will retain the same record interface, source observations, action history and model. A retrieval baseline will reconstruct a task summary and proposals from the available records. The comparison mechanism will retain explicit links between agreements and actions, flag affected links after a registered change, and propose local updates. Both can ask questions and access the same evidence. Model configuration and resource budgets will be matched as closely as practicable, with actual usage reported.

Matched change opportunities will support a comparison of the first proposal after each change. The primary measure will be invalid carryover: a proposal presented as current and executable still depends on an object, relation or goal invalidated by that change. Waiting-for-clarification proposals will be recorded separately rather than counted as correct actions. Resumption time, unnecessary revision of valid work, task quality and supervision effort will reveal trade-offs. Counterbalanced task sequences will reduce order effects; estimates will remain exploratory if the sample does not support a precise comparison.

What the assistant should remember

The first memory design will retain object references, agreed relations, rejected alternatives and their stated reasons, with timestamps and revision status. For example, a previous arrangement may have been rejected because it blocked a viewing line. On return, the assistant should distinguish that reason from the old coordinates and check whether it still applies. Participants can inspect, correct and remove records. Temporary decisions will not automatically become enduring personal preferences.

Interviews will examine continuity, familiarity and whether participants feel understood. These experiences will be analysed alongside memory errors and actual task outcomes. Three visits can establish evidence about early reuse, not a durable personal relationship. Stable personality, cross-device continuity and broader daily-life support remain longer-term research opportunities.

Technical checks before deployment

Offline sequences will test ambiguous references, moved objects, superseded instructions, incomplete evidence and scene changes during model response. Expected outcomes will be independently annotated, allowing multiple acceptable interpretations. Development and test cases will be separated by scene and user where possible. Failure analysis will distinguish sensing, interpretation, stale dependencies and execution. If tracking or language-model reliability is inadequate, the study will retain known scene state and narrow the action vocabulary rather than rely on hidden human intervention.

Expected contributions and research fit

The programme aims to contribute an empirical account of how speech, drawing and manipulation establish and revise shared spatial intentions; an implemented interaction mechanism connecting these intentions to inspectable actions; and comparative evidence about repair and continuation as goals and scenes change. A useful outcome may also identify conditions where ordinary conversation and previews are sufficient, avoiding unnecessary interface complexity.

This direction connects to Associate Professor Türkay's work through embodied conversational interaction [1], drawing-supported spatial dialogue [2] and the integration of immersive tools into authentic learning activity [7]. It also aligns with the advertised QUT topic on intelligent agents in immersive systems, which includes games, learning and complex data analysis [8]. I would welcome discussion of how the testbed could complement CLEVAR's ongoing projects. Exhibition layout is my proposed starting context; geological data exploration or learning tasks would require domain collaboration and suitable measures.

Recent work further constrains the contribution. Eye2Eye already addresses revisable common ground [6], while SpatialPrompt turns sketches and voice into constraints for 3D generation [9]. SpatiaLLM uses constraint-verified spatial memory for adaptive placement [10]. The research therefore tests selective updating of joint spatial work and its interaction costs, rather than claiming that multimodal input, editable memory or spatial constraints are new in themselves. These comparisons will be refined through continued literature review and prototype development.

Preparation and feasibility

My background combines architectural training, immersive exhibition delivery and computer science. Work on CAVE projection and interaction gives me experience with spatial layouts and on-site system integration. Game Fellow prototypes and research writing motivate my interest in an AI that contributes to joint activity; participant evaluation remains planned. My image-detection capstone provides experience with reproducible output audits and analysing errors separately from aggregate scores. I will build Unity/C# and HCI study skills as part of the doctoral programme.

Year 1 will complete formative work, develop XR implementation skills and establish an end-to-end prototype. Year 2 will develop and evaluate the working record and repair workflow. Year 3 will examine repeated use and change, followed by synthesis and necessary follow-up in Year 4. The project reuses existing models, a small asset library and one application. Equipment, recruitment and the final duration will be agreed with the supervisory team.

Research ethics

Ethics approval will precede recruitment. Recording will be limited to task-relevant activity with informed consent. Participants can pause recording, take breaks and withdraw. Any externally processed speech or scene information will be disclosed, and unrelated personal material will be excluded. Study error manipulations and Wizard-of-Oz behaviour will be disclosed and logged. The system will offer undo and a way to inspect or remove retained task history. Proposed studies do not require clinical diagnosis or make claims about emotional dependence.

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